Aer simulator fails to work with custom parametrized circuits and VQE

https://github.com/Qiskit/qiskit/issues/10589

ROW 75

Aer simulator fails to work with custom parametrized circuits and VQE #10589

New issue

Closed

Gaidailgor opened on Aug 8, 2023

Informations

- **Qiskit Aer version**: 0.12.2
- **Python version**: 3.11.3
- **Operating system**: Windows 10

What is the current behavior?

When I try to execute the code below, I get the following error:

```
Traceback (most recent call last):
  File "C:\Users\User\Desktop\python3.11_venv\Lib\site-packages\qiskit\extensions\hamiltonian_gate.py", line 9
    return scipy.linalg.expm(-1j * self.params[0] * float(self.params[1]))
                                                    ^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\User\Desktop\python3.11_venv\Lib\site-packages\qiskit\circuit\parameterexpression.py", line 4
    raise TypeError(
TypeError: ParameterExpression with unbound parameters ({ParameterVectorElement(angles[0])}) cannot be cast to

The above exception was the direct cause of the following exception:

Traceback (most recent call last):
  File "C:\Users\User\Desktop\python3.11_venv\Lib\site-packages\qiskit\transpiler\passes\basis\unroll_custom_d
    unrolled = getattr(node.op, "definition", None)
               ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\User\Desktop\python3.11_venv\Lib\site-packages\qiskit\circuit\instruction.py", line 239, in d
    self._define()
  File "C:\Users\User\Desktop\python3.11_venv\Lib\site-packages\qiskit\extensions\hamiltonian_gate.py", line 1
    qc._append(UnitaryGate(self.to_matrix()), q[:], [])
                           ^^^^^^^^^^^^^^^
  File "C:\Users\User\Desktop\python3.11_venv\Lib\site-packages\qiskit\circuit\gate.py", line 53, in to_matrix
    return self.__array__(dtype=complex)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\User\Desktop\python3.11_venv\Lib\site-packages\qiskit\extensions\hamiltonian_gate.py", line 9
    raise TypeError(
TypeError: Unable to generate Unitary matrix for unbound t parameter angles[0]

The above exception was the direct cause of the following exception:

Traceback (most recent call last):
  File "C:\Users\User\Desktop\python3.11_venv\Lib\site-packages\qiskit_algorithms\minimum_eigensolvers\vqe.py"
    estimator_result = job.result()
                       ^^^^^^^^^^^^
  File "C:\Users\User\Desktop\python3.11_venv\Lib\site-packages\qiskit\primitives\primitive_job.py", line 55,
    return self._future.result()
           ^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\User\AppData\Local\Programs\Python\Python311\Lib\concurrent\futures\_base.py", line 456, in r
    return self.__get_result()
           ^^^^^^^^^^^^^^^^^^^
  File "C:\Users\User\AppData\Local\Programs\Python\Python311\Lib\concurrent\futures\_base.py", line 401, in _
    raise self._exception
  File "C:\Users\User\AppData\Local\Programs\Python\Python311\Lib\concurrent\futures\thread.py", line 58, in r
    result = self.fn(*self.args, **self.kwargs)
             ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\User\Desktop\python3.11_venv\Lib\site-packages\qiskit_aer\primitives\estimator.py", line 128,
    return self._compute(circuits, observables, parameter_values, run_options)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\User\Desktop\python3.11_venv\Lib\site-packages\qiskit_aer\primitives\estimator.py", line 188,
    self._transpile_circuits(circuits)
  File "C:\Users\User\Desktop\python3.11_venv\Lib\site-packages\qiskit_aer\primitives\estimator.py", line 481,
    circuit = self._transpile(circuit)
              ^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\User\Desktop\python3.11_venv\Lib\site-packages\qiskit_aer\primitives\estimator.py", line 468,
    return transpile(circuits, self._backend, **self._transpile_options.__dict__)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\User\Desktop\python3.11_venv\Lib\site-packages\qiskit\compiler\transpiler.py", line 418, in t
    out_circuits = pm.run(circuits, callback=callback)
                   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

Assignees
No one assigned

Labels
bug

Type
No type

Projects
No projects

Milestone
No milestone

Relationships
None yet

Development
No branches or pull requests

Notifications Customize

Subscribe

You're not receiving notifications from this thread.

Participants

```
    new_dag = pass_.run(dag)
              ^^^^^^^^^^^^^^
  File "C:\Users\User\Desktop\python3.11_venv\Lib\site-packages\qiskit\transpiler\passes\basis\unroll_custom_d
    raise QiskitError(f"Error decomposing node {node.name}: {err}") from err
qiskit.exceptions.QiskitError: 'Error decomposing node hamiltonian: Unable to generate Unitary matrix for unbo

The above exception was the direct cause of the following exception:

Traceback (most recent call last):
  File "C:\Users\User\Desktop\MA-QAOA\temp\my_test.py", line 29, in <module>
    test_vqe()
  File "C:\Users\User\Desktop\MA-QAOA\temp\my_test.py", line 25, in test_vqe
    result = vqe.compute_minimum_eigenvalue(-hamiltonian)
             ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\User\Desktop\python3.11_venv\Lib\site-packages\qiskit_algorithms\minimum_eigensolvers\vqe.py"
    optimizer_result = self.optimizer(
                       ^^^^^^^^^^^^^^^
  File "C:\Users\User\Desktop\python3.11_venv\Lib\site-packages\scipy\optimize\_minimize.py", line 710, in mir
    res = _minimize_lbfgsb(fun, x0, args, jac, bounds,
          ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\User\Desktop\python3.11_venv\Lib\site-packages\scipy\optimize\_lbfgsb_py.py", line 307, in _m
    sf = _prepare_scalar_function(fun, x0, jac=jac, args=args, epsilon=eps,
         ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\User\Desktop\python3.11_venv\Lib\site-packages\scipy\optimize\_optimize.py", line 383, in _pr
    sf = ScalarFunction(fun, x0, args, grad, hess,
         ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\User\Desktop\python3.11_venv\Lib\site-packages\scipy\optimize\_differentiable_functions.py",
    self._update_fun()
  File "C:\Users\User\Desktop\python3.11_venv\Lib\site-packages\scipy\optimize\_differentiable_functions.py",
    self._update_fun_impl()
  File "C:\Users\User\Desktop\python3.11_venv\Lib\site-packages\scipy\optimize\_differentiable_functions.py",
    self.f = fun_wrapped(self.x)
             ^^^^^^^^^^^^^^^^^^^
  File "C:\Users\User\Desktop\python3.11_venv\Lib\site-packages\scipy\optimize\_differentiable_functions.py",
    fx = fun(np.copy(x), *args)
         ^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\User\Desktop\python3.11_venv\Lib\site-packages\qiskit_algorithms\minimum_eigensolvers\vqe.py"
    raise AlgorithmError("The primitive job to evaluate the energy failed!") from exc
qiskit_algorithms.exceptions.AlgorithmError: 'The primitive job to evaluate the energy failed!'
```

Apparently, parametrized Hamiltonian gates fail to convert to unitaries when `AerEstimator` is used. However, if `qiskit.primitives.Estimator` is used instead, then everything works as expected. As far as I understand, one should be able to use these estimators interchangeably.

Steps to reproduce the problem

The following code reproduces the problem:

```
from qiskit import QuantumCircuit
from qiskit.circuit import ParameterVector
from qiskit.primitives import Estimator
from qiskit_aer.primitives import Estimator as AerEstimator
from qiskit.quantum_info import SparsePauliOp, Pauli
from qiskit_algorithms.minimum_eigensolvers import VQE
from scipy import optimize


def get_custom_ansatz(num_qubits: int) -> QuantumCircuit:
    params = ParameterVector('angles', num_qubits)
    circuit = QuantumCircuit(num_qubits)
    for i in range(num_qubits):
        circuit.hamiltonian(Pauli('X'), params[i], [i])
    return circuit


def test_vqe():
    n = 12
    estimator = AerEstimator()  # Estimator() works
    hamiltonian = SparsePauliOp.from_sparse_list([('ZZ', [0, 1], -1)], n)
    ansatz = get_custom_ansatz(n)
    optimizer = optimize.minimize
    vqe = VQE(estimator, ansatz, optimizer)
    result = vqe.compute_minimum_eigenvalue(-hamiltonian)
    return -result.eigenvalue.real


test_vqe()
```

What is the expected behavior?

I expect no errors.

Suggested solutions

Gaidailgor added bug on Aug 8, 2023

mtreinish transferred this issue from Qiskit/qiskit-aer on Aug 8, 2023

ikkoham on Aug 8, 2023 — Contributor

Thanks. As @mtreinish transferred, this is the transpilation issue.

minimal code to reproduce this is

```
from qiskit import QuantumCircuit, transpile
from qiskit.circuit import ParameterVector
from qiskit_aer import AerSimulator


def get_custom_ansatz(num_qubits: int) -> QuantumCircuit:
    params = ParameterVector('angles', num_qubits)
    circuit = QuantumCircuit(num_qubits)
    for i in range(num_qubits):
        circuit.hamiltonian(Pauli('X'), params[i], [i])
    return circuit


transpile(get_custom_ansatz(n), AerSimulator())
```

mtreinish on Aug 8, 2023 — Member

It's actually more of an issue with the implementation of the `HamiltonianGate`, it is raising errors when there is an unbound parameter when the transpiler is asking for it's definition to unroll it into terms that can be used for basis translation. There isn't a way to transpile it directly with an unbound parameter because it's definition can only be realized with a numeric parameter for `time` because the definition just returns a unitary gate, and we can't define the matrix without a numerical value for `time`.

Cryoris on Aug 15, 2023 — Contributor

Yeah this only work with the `Estimator` in Qiskit (Terra) as it binds the parameter values before simulating the circuit.

If you'd like to use operator evolutions with the Aer Estimator, you can use the `PauliEvolutionGate` instead of the `QuantumCircuit.hamiltonian` function. The `PauliEvolutionGate` implements a Trotter expansion, so it is not exact, but this is what we would also have to run on a quantum computer.

I'll go ahead and close this issue, but feel free to re-open if the answers above didn't answer your question.

Cryoris closed this as completed on Aug 15, 2023